Reflective AI Feedback for Personal Productivity

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I have used task lists and productivity systems for most of my academic life, ranging from handwritten notebooks to modern digital productivity platforms. While these tools are effective at recording tasks and reminders, they often fail to help users understand how they plan, execute, and reflect on their work over time. In my own experience, I frequently created overly ambitious task lists, leading to frustration when goals were not completed. Additionally, most productivity tools offer limited insight into long-term behavioral patterns such as task overload, inconsistent pacing, or changes in motivation throughout the week.

Recent advances in artificial intelligence, particularly large language models (LLMs), present an opportunity to move beyond static task tracking toward reflective analysis of human workflows. Prior research has shown that AI systems can support human thinking by assisting with ideation, reflection, and decision-making rather than replacing human agency [2] . Similarly, work on human–AI collaboration demonstrates that AI is most effective when positioned as a cognitive partner that provides feedback and perspective [**?**]. These findings motivate the design of productivity tools that emphasize reflection and insight rather than automation alone.

This project proposes the design and implementation of a local, privacy-preserving productivity application augmented with an AI-based reflection system. The core research question guiding this work is: *Does reflective feedback from an AI system influence user productivity behavior and planning habits over time?* Rather than directly executing tasks on behalf of the user, the AI will analyze task metadata and daily reflections to generate summaries, observations, and personalized suggestions regarding productivity patterns. This approach aligns with the concept of "supermind intelligence," which describes systems that enhance collective or individual problem-solving through structured feedback and insight rather than autonomous control [1].

The software will allow users to organize tasks by day of the week using a visual interface inspired by timeline-based productivity tools. Each task will include a title, description, creation time, and completion status. Users may optionally write short daily reflections describing their perceived productivity or challenges. The AI system, running locally via an open-source language model, will have access to this structured data and will periodically generate reflective feedback. Examples include identifying periods of over-planning, changes in task completion rates, or correlations between reflections and productivity outcomes. To evaluate the effectiveness of reflective AI feedback, the project will involve a small user study.

While the scope of the study will be limited, it will provide initial evidence on whether AI-driven reflection can meaningfully support personal productivity. By focusing on reflection, transparency, and user agency, the system aims to support users in understanding their own work habits.

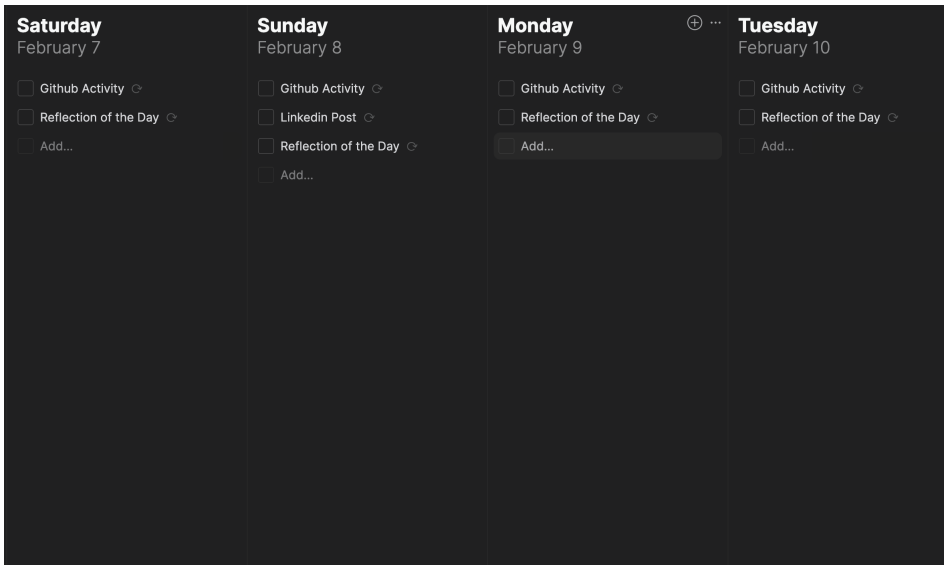

Figure 1: Main Page Interface

Figure 2: Report Panel Draft

Appendix

Planned features:

- User can create tasks assigned to a specific day of the week.
- Each task includes a title, optional description, creation timestamp, and completion status.
- User can mark tasks as completed or incomplete.
- User can edit or delete existing tasks.
- User can view a weekly timeline showing tasks organized by day.
- User can write a short daily reflection associated with each day.
- The system stores task and reflection data locally on the users machine.
- A local AI model analyzes task metadata and reflections to generate weekly productivity summaries.
- The AI provides observations about task load, completion trends, and pacing across the week.
- The AI suggests actionable productivity tips based on observed patterns.
- User can view historical AI feedback from previous weeks.
- (Stretch Goal) Task classification by type: academic, personal, work related.
- (Stretch Goal) Productivity metrics dashboard.
- (Stretch Goal) Task completion streak.

References

[1] Heyman, J. L., Rick, S. R., Giacomelli, G., Wen, H., Laubacher, R., Taubenslag, N., Knicker, M., Jeddi, Y., Ragupathy, P., Curhan, J., and Malone, T. Supermind ideator: How scaffolding human-ai collaboration can increase creativity. In *Proceedings of the ACM Collective Intelligence Conference* (New York, NY, USA, 2024), CI '24, Association for Computing Machinery, p. 18–28.

[2] Reicherts, L., Zhang, Z. T., von Oswald, E., Liu, Y., Rogers, Y., and Hassib, M. AI, help me think—but for myself: Assisting people in complex decision-making by providing different kinds of cognitive support. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (New York, NY, USA, 2025), CHI '25, Association for Computing Machinery.